

TRACKAI · SIH 26169 — FSOC Performance Report

AI-based virtual camera tracking and coarse/fine optical alignment — summary for review

Key Performance Indicators — Simulated Scenario Mode

Sessions run:	280
Samples logged:	8353
Average total error:	1.369°
Fine-lock sample rate:	78.6%

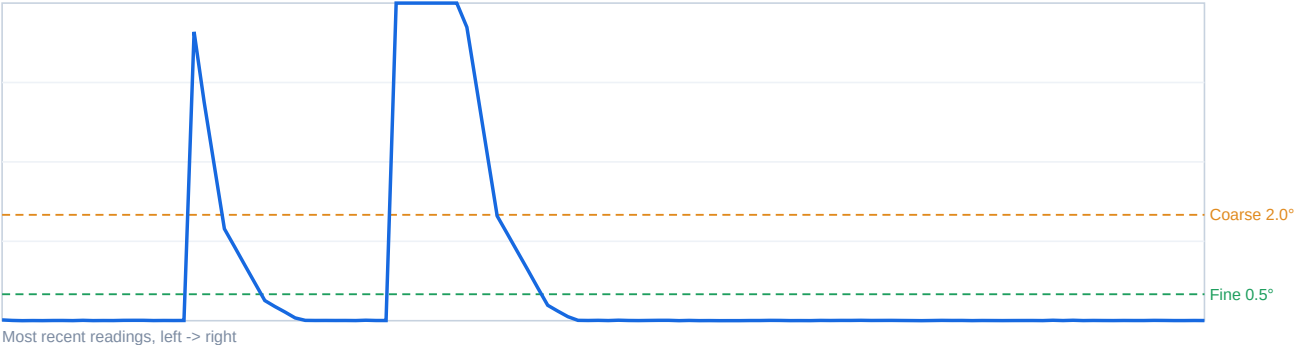
Alignment Workflow

- AI detection -> target lock -> H/V error estimation -> LEFT/RIGHT/UP/DOWN guidance
- Coarse alignment gate $\leq 2.0^\circ$ -> fine alignment -> automatic fine lock $\leq 0.5^\circ$
- All tracking readings, commands, alerts and events are persisted in SQLite.
- A second, independent pipeline (below) runs real OpenCV/YOLO computer-vision detection on a live camera feed.

Before vs. After Alignment — Simulated Scenario

METRIC	BEFORE (ACQUISITION)	AFTER (LOCKED)	CHANGE
Total angular error (°)	5.99°	0.00°	↓ 5.99°
AI confidence (%)	90.7%	98.0%	↑ 7.3%

Total Angular Error Over Session (simulated)



Live AI Camera Pipeline

Real OpenCV / YOLO computer-vision detection results — not simulated

Active detector: OPENCV · Camera source: SYNTHETIC · Logged detections: 400233

Before vs. After Alignment — Live Camera

METRIC	BEFORE (ACQUISITION)	AFTER (LOCKED)	CHANGE
Total angular error (°)	8.09°	0.51°	↓ 7.58°
AI confidence (%)	97.4%	97.4%	→ 0.0%

Total Angular Error Over Time (live camera)



Free-Space Optical / Inter-Satellite Laser Link

Key technical challenges and this system's mitigation approach

Pointing / tracking error

Beam divergence over long ranges means even sub-mrad mispointing misses the receiver aperture entirely.

Mitigation: Closed-loop coarse->fine gated tracking (this system's core function): coarse acquisition $\leq 2^\circ$ then fine lock $\leq 0.5^\circ$, continuously re-corrected.

Platform vibration & micro-jitter

Satellite bus vibration, reaction wheels and thermal snap induce continuous angular jitter on the optical bench.

Mitigation: Fast inner control loop (fine-steering mirror / gimbal) with bandwidth well above the disturbance spectrum; vibration isolation mounts.

Atmospheric turbulence & scintillation

For ground-to-space or air-to-ground links, turbulence causes beam wander, scintillation fading and angle-of-arrival noise.

Mitigation: Adaptive optics, spatial diversity (multiple receive apertures), and larger link margin budgeted for fade statistics.

Acquisition time / re-acquisition after occlusion

Long search times after signal loss (occlusion, maneuver, fade) reduce effective link availability.

Mitigation: Wide-FOV coarse search pattern with fast beacon detection (this system's SEARCHING→ACQUISITION phase), predictive pointing from orbit ephemeris.

Beam divergence vs. link budget trade-off

Narrower beams need less power but demand tighter pointing; wider beams relax pointing but lose power density.

Mitigation: System-level trade study balancing pointing accuracy achievable against required divergence and transmit power.

Relative velocity / Doppler & lead-ahead angle

Fast relative motion between satellites requires transmitting slightly ahead of the receiver's true position.

Mitigation: Orbit-propagator-driven lead-angle compensation feeding the pointing controller in addition to the vision-based correction.

Thermal-optical drift

Thermal cycling in orbit shifts optical alignment (boresight drift) between the tracking sensor and the transmit laser.

Mitigation: Periodic boresight calibration cycles and temperature-compensated optical mounts.

Background / solar interference

Sunlit backgrounds or stray light can saturate the tracking sensor or mimic a bright beacon.

Mitigation: Narrowband spectral filtering matched to the beacon wavelength, plus the brightness/circularity confidence heuristic used by this system's detector.

Clock synchronization & timing jitter

Coarse pointing windows and beacon scheduling need tight time sync between terminals.

Mitigation: GNSS-disciplined or inter-satellite time-transfer synchronization with margin built into the acquisition window.